

Supplementary Winter Examination – 2025

Course: B. Tech.

Branch: Civil Engineering

Semester: II

Civil Engineering

Subject Code & Name: 23UD1191ES106 Basic Civil Engineering

Max Marks: 60

Date: 24/01/26

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	Which property of bricks mainly affects their suitability in load bearing walls?	CO1	1				
	<table border="1"> <tr> <td>A) Colour</td><td>B) Compressive strength</td><td>C) Size uniformity</td><td>D) Efflorescence</td></tr> </table>	A) Colour	B) Compressive strength	C) Size uniformity	D) Efflorescence		
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2	Timber used in construction should preferably be seasoned to reduce	CO1	1				
	<table border="1"> <tr> <td>A) Strength</td><td>B) Durability</td><td>C) Moisture content</td><td>D) Density</td></tr> </table>	A) Strength	B) Durability	C) Moisture content	D) Density		
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3	The main function of sand in mortar is to	CO1	1				
	<table border="1"> <tr> <td>A) Increase setting time</td><td>B) Reduce shrinkage</td><td>C) Improve binding</td><td>D) Increase strength of cement</td></tr> </table>	A) Increase setting time	B) Reduce shrinkage	C) Improve binding	D) Increase strength of cement		
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4	The primary function of foundation in a building is to	CO2	1				
	<table border="1"> <tr> <td>A) Beautify structure</td><td>B) Transfer load safely to soil</td><td>C) Reduce cost</td><td>D) Increase height</td></tr> </table>	A) Beautify structure	B) Transfer load safely to soil	C) Reduce cost	D) Increase height		
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5	Isolated footing is generally suitable when	CO2	1				

	A) Soil has low bearing capacity	B) Columns are closely spaced	C) Loads are moderate	D) Water table is high		
6	Plinth level of a building is mainly provided to				CO2	1
	A) Improve appearance	B) Prevent dampness	C) Reduce load	D) Increase ventilation		
7	Lintels are provided over openings mainly to				CO3	1
	A) Support roof	B) Transfer wall load	C) Improve lighting	D) Reduce span of slab		
8	Symbols used in building drawings help to				CO3	1
	A) Increase drawing size	B) Reduce construction cost	C) Convey information clearly	D) Avoid site supervision		
9	The main objective of environmental engineering is to				CO4	1
	A) Increase industrial growth	B) Protect public health	C) Maximize resource use	D) Improve aesthetics		
10	Primary treatment of sewage mainly involves				CO4	1
	A) Biological process	B) Chemical treatment	C) Physical removal of solids	D) Disinfection		
11	Solid waste management aims at				CO5	1
	A) Waste generation	B) Safe disposal	C) Increasing waste	D) Ignoring segregation		
12	Flexible pavements are those which				CO5	1
	A) Have cement concrete surface	B) Distribute load through grain action	C) Are rigid in nature	D) Do not deform		
Q. 2	Solve the following.					12
A)	Explain basic engineering properties and uses of materials - brick				CO1	6
B)	Explain Seasoning of timber				CO1	6

Q.3	Solve the following.		12
A)	What are the different components of a building? Explain any 5 of them.	CO2	6
B)	What is plan ,elevation and section in building drawing ?	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Explain Water Treatment Process in detail.	CO3	6
B)	What is Process of Environmental Impact Assessment?	CO3	6
C)	Write short note on Solid Waste Management.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain Components of Pavement: Sub-grade, Sub-Base, Base-course, Wearing course.	CO4	6
B)	Write short note on Classification of Roads	CO4	6
C)	Explain Road Safety measures in detail.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	What are the principles of survey?	CO5	6
B)	Explain Plane table surveying.	CO5	6
C)	What is Compass Surveying? Explain its principle.	CO5	6
	*** End ***		